

A Structural Pandrosion Reduction for Smale's Mean Value Conjecture

The Inverse Quotient and the Fiber Vanishing Identity

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Abstract

We develop a framework for Smale's mean value conjecture based on the *Pandrosion inverse*: the reverse evaluation of the Pandrosion quotient from a critical point to the origin. This yields three observations. First, the Smale ratio decomposes as $S(c) = |c| \cdot \prod |z_j|$ where z_j are the co-fiber points of the critical value. Second, we record the *fiber vanishing identity* $\sum 1/((\alpha_j - c)^2 P'(\alpha_j)) = 0$ and the *fiber-bridge* $\sum 1/((\alpha_j - c) P'(\alpha_j)) = -1/P(c)$, which together define an orthogonality structure $\mathbf{u} \perp \mathbf{v}$, $\mathbf{u} \perp 1/\mathbf{v}$ in $\mathbb{C}^d$. Third, we record the *Pandrosion reduction*: at each critical point, $R(c) = \tilde{q}(c)/d$ where $\tilde{q}(0) = d - 1$, reducing Smale's conjecture to the statement that $\tilde{q}$ does not exceed its value at the origin at every root of P' . For $d = 3$ we give a closed algebraic proof in the Pandrosion normal form; for general d we give only a first-order (linear) perturbation result at the extremal $z^d - z$, which does not extend to a proof for $d \geq 4$.

Scope statement. The results of this paper are a structural reduction, not a proof of Smale's MVC for $d \geq 4$. The $d = 3$ section is now a closed algebraic proof in the normal form $P(z) = z^3 + bz^2 + z$. The “perturbation proof for all d ” of Section 6 below is only a first-order (linear) stability statement, not a full local maximum, and certainly not a global bound. Readers seeking the sharper (but still incomplete) $d = 5$ analysis should consult Part VI.

Reader's guide. This paper is the hinge of the MVC branch. Parts II–III expressed the conjecture in quotient language and derived root-side constraints; here the critical value is studied through its fiber. The output is the reduction $R(c) = \tilde{q}(c)/d$, which is the exact algebraic form used in the quartic paper (Part V) and the centered quintic program (Part VI). The perturbation section should be read as local evidence around the extremal family, not as a global argument.

1 Setup and notation

Let $P(z) = zR(z)$, monic of degree d , with $R(0) = 1$ (equivalently $P'(0) = 1$). The roots of P are $\alpha_0 = 0, \alpha_1, \dots, \alpha_{d-1}$; the critical points (roots of P') are $c_1, \dots, c_{d-1}$. The Smale ratio is $S(c) = |R(c)| = |P(c)/c|$.

2 The Pandrosion inverse

Theorem 2.1 (Pandrosion inverse = Smale ratio). $Q(c, 0) = P(c)/c$, hence $S(c) = |Q(c, 0)|$.

Since $P'(c) = Q(c, c) = 0$, the quotient factors as $Q(c, z) = (z - c)Q_2(c, z)$, where $Q_2(c, z) = (P(z) - P(c))/(z - c)^2$ is monic of degree $d - 2$.

Definition 2.2. The co-fiber of a critical point c is $\text{Fib}(c) = \{z_1, \dots, z_{d-2}\}$, the roots of $Q_2(c, \cdot)$, i.e., the solutions of $P(z) = P(c)$ other than $z = c$ (with multiplicity).

Theorem 2.3 (Co-fiber product).

$$S(c) = |c| \cdot \prod_{j=1}^{d-2} |z_j|. \quad (1)$$

Proof. $Q_2(c, 0) = -P(c)/c^2$, and since Q_2 is monic, $|Q_2(c, 0)| = \prod |z_j|$. $\square$

3 The three-identity system

Theorem 3.1 (Root vanishing — classical Lagrange–Sylvester identity). For any polynomial P of degree $d \geq 2$ with simple roots $\alpha_0, \dots, \alpha_{d-1}$,

$$\sum_{j=0}^{d-1} \frac{1}{P'(\alpha_j)} = 0. \quad (2)$$

This is the partial-fraction identity for $1/P(z)$ in the limit $z \rightarrow \infty$ (the coefficient of $1/z$ in the Laurent expansion of $1/P$ vanishes when $\deg P \geq 2$); see Markushevich [8], Vol. I, Ch. 5. We restate it here in Pandrosion notation for the use of §4.

Theorem 3.2 (Fiber-bridge). At each critical point c :

$$\boxed{\sum_{j=0}^{d-1} \frac{1}{(\alpha_j - c) P'(\alpha_j)} = -\frac{1}{P(c)}}. \quad (3)$$

Proof. Let $u_j = 1/((\alpha_j - c)P'(\alpha_j))$. From the root vanishing: $\sum u_j(\alpha_j - c) = 0$, so $\sum u_j \alpha_j = c \sum u_j$. Since $\sum \alpha_j u_j = -c/P(c)$ (the bridge formula for $z/P(z)$ at $z = c$), we get $\sum u_j = -1/P(c)$. $\square$

Theorem 3.3 (Fiber vanishing). At each critical c :

$$\boxed{\sum_{j=0}^{d-1} \frac{1}{(\alpha_j - c)^2 P'(\alpha_j)} = 0}. \quad (4)$$

Proof. $Q_2(c, \alpha_j) = -P(c)/(\alpha_j - c)^2$. Since $\deg Q_2 = d - 2 < d - 1$, the Lagrange identity $\sum Q_2(c, \alpha_j)/P'(\alpha_j) = 0$ gives the result after dividing by $-P(c)$. $\square$

Proposition 3.4 (Orthogonality). With $u_j = 1/((\alpha_j - c)P'(\alpha_j))$ and $v_j = \alpha_j - c$:

$$\mathbf{u} \cdot \mathbf{v} = 0, \quad \mathbf{u} \cdot (1/\mathbf{v}) = 0, \quad |\langle \mathbf{u}, \mathbf{1} \rangle| = \frac{1}{S(c)|c|}. \quad (5)$$

4 The Pandrosion reduction

Theorem 4.1 (Pandrosion reduction). At each critical point c (where $P'(c) = 0$):

$$R(c) = \frac{\tilde{q}(c)}{d}, \quad \tilde{q}(z) = \sum_{k=1}^{d-1} k a_{d-k} z^{d-1-k}, \quad (6)$$

where $P(z) = z^d + a_{d-1}z^{d-1} + \dots + a_1z$. In particular, $\tilde{q}(0) = (d-1)a_1 = d-1$.

Proof. From $P'(c) = 0$: $dc^{d-1} = -\sum_{k=1}^{d-1} ka_{d-k}c^{d-1-k} + d \cdot c^{d-1} \dots$. More directly: $R(c) = c^{d-1} + a_{d-1}c^{d-2} + \dots + 1$, and using $c^{d-1} = -(1/d) \sum_{k=1}^{d-1} ka_{d-k}c^{d-1-k}$ from the critical equation:

$$R(c) = \frac{1}{d}(a_{d-1}c^{d-2} + 2a_{d-2}c^{d-3} + \dots + (d-1)) = \frac{\tilde{q}(c)}{d}. \quad \square$$

Corollary 4.2 (Reformulation of Smale). *Smale's conjecture is equivalent to: for all choices of $a_1 = 1, a_2, \dots, a_{d-1} \in \mathbb{C}$, there exists a root c of $P'(z)/d$ such that $|\tilde{q}(c)| \leq d-1 = \tilde{q}(0)$.*

Remark 4.3. *At the extremal $P = z^d - z$: all $a_k = 0$ except $a_1 = -1$ (with our sign convention, $R(z) = z^{d-1} - 1$, adjusting normalisation). Then $\tilde{q}$ is constant and $S = (d-1)/d$ for all critical points. The numerical evidence suggests that non-trivial coefficient perturbations lower the minimum, but this remark is not used as a proof for $d \geq 4$.*

5 Proof for $d = 3$

Theorem 5.1. *Smale's conjecture holds for $d = 3$, with equality only for $P(z) = z^3 - z$ (up to rotation).*

Proof. Write $P(z) = z^3 + bz^2 + z$, so $R(z) = z^2 + bz + 1$ and $P'(z) = 3z^2 + 2bz + 1$. At a critical point c ,

$$3c^2 + 2bc + 1 = 0, \quad R(c) = c^2 + bc + 1 = \frac{bc + 2}{3}.$$

If $b = 0$, both critical points give $|R(c)| = 2/3$.

Assume $b \neq 0$ and set $t_j = bc_j + 2$ for the two critical points. Substituting $c = (t-2)/b$ into the critical equation gives

$$A(t-2)^2 + 2t - 3 = 0, \quad A := \frac{3}{b^2}.$$

Thus $t_1 + t_2 = 4 - 2/A$ and $t_1 t_2 = 4 - 3/A$; eliminating A gives

$$2t_1 t_2 - 3(t_1 + t_2) + 4 = 0, \quad \text{or} \quad (2t_1 - 3)(2t_2 - 3) = 1.$$

If both $|t_1|$ and $|t_2|$ were strictly larger than 2, then both $|2t_j - 3|$ would be strictly larger than 1 (the point 3 is at distance 1 from the circle $|2t| = 4$). Their product would then have modulus larger than 1, contradicting $(2t_1 - 3)(2t_2 - 3) = 1$. Therefore some j has $|t_j| \leq 2$, and hence

$$\min_j |R(c_j)| = \frac{1}{3} \min_j |t_j| \leq \frac{2}{3}.$$

Equality occurs at $b = 0$, i.e. at $P(z) = z^3 + z$ in this normalisation; after rotation this is the usual extremal family. $\square$

6 First-order perturbation stability for general d

Proposition 6.1 (First-order stability at the extremal — not a global proof). *At the extremal $P = z^d - z$, the bound $\min S \leq (d-1)/d$ is stable under first-order perturbations: there is no tangent direction in parameter space along which all critical values strictly exceed $(d-1)/d$ at first order. This is a linear-in- ε statement and does not imply a strict local maximum, let alone a global bound.*

Proof. At the extremal, critical points are $c_k = d^{-1/(d-1)}\omega^k$ where $\omega = e^{2\pi i/(d-1)}$. Write $\tilde{q}(z) = (d-1) + z r(z)$ for a perturbation r of degree $d-3$. Then:

$$\sum_{k=1}^{d-1} c_k r(c_k) = \sum_m r_m \sum_k c_k^{m+1} = \sum_m r_m \rho^{m+1} \sum_k \omega^{k(m+1)} = 0,$$

since $\sum_k \omega^{k(m+1)} = 0$ for $1 \leq m+1 \leq d-2$. Therefore, the $d-1$ values $\{c_k r(c_k)\}$ cannot all have positive real part, so there exists k with $\operatorname{Re}(c_k r(c_k)) \leq 0$, giving $|\tilde{q}(c_k)| \leq d-1 + O(\|r\|^2)$. $\square$

7 Numerical verification

d	$\sup \min S / (d-1)/d$	Tests	Status
3	1.0000	200k	Proved
4	0.9784	20k	Verified
5	0.9194	20k	Verified
6	0.8828	20k	Verified
7	0.8779	20k	Verified
8	0.8343	20k	Verified

Zero violations were observed across all tested degrees. The statement that the extremal family is the unique global maximiser remains empirical for $d \geq 4$ in this note.

8 Conclusion

The Pandrosion inverse reads Smale's conjecture as a statement about the branched covering $z \mapsto P(z)$: the Smale ratio equals the product of co-fiber distances (Theorem 2.3), and is controlled by the polynomial $\tilde{q}$ of degree $d-2$ whose value at the origin is exactly $d-1$ (Theorem 4.1).

The conjecture reduces to showing that this degree- $(d-2)$ polynomial cannot exceed its value at the origin at *every* root of its companion polynomial P'/d of degree $d-1$.

The first-order perturbation result of Section 6 is a linear stability statement only; the full coupling between $\tilde{q}$ and P'/d through the shared coefficients $a_2, \dots, a_{d-1}$ is what remains unresolved, and what blocks a proof for $d \geq 4$ from this angle.

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